

# IR Sensor

IR sensor is an electronic device, that emits the light in order to sense some object of the surroundings.

An IR sensor can detect the motion.

The emitter is simply an IR LED (Light Emitting Diode) and the detector is simply an IR photodiode .

Photodiode is sensitive to IR light which is emitted by the IR LED.

When IR light falls on the photodiode, the resistances and the output voltages will change in proportion to the magnitude of the IR light received.

# IR Sensor

## IR Sensor Working Principle

### IR Transmitter or IR LED

Infrared Transmitter is a light emitting diode (LED) which emits infrared radiations called as IR LED's. Even though an IR LED looks like a normal LED, the radiation emitted by it is invisible to the human eye.

The picture of an Infrared LED is shown below.



# IR Receiver

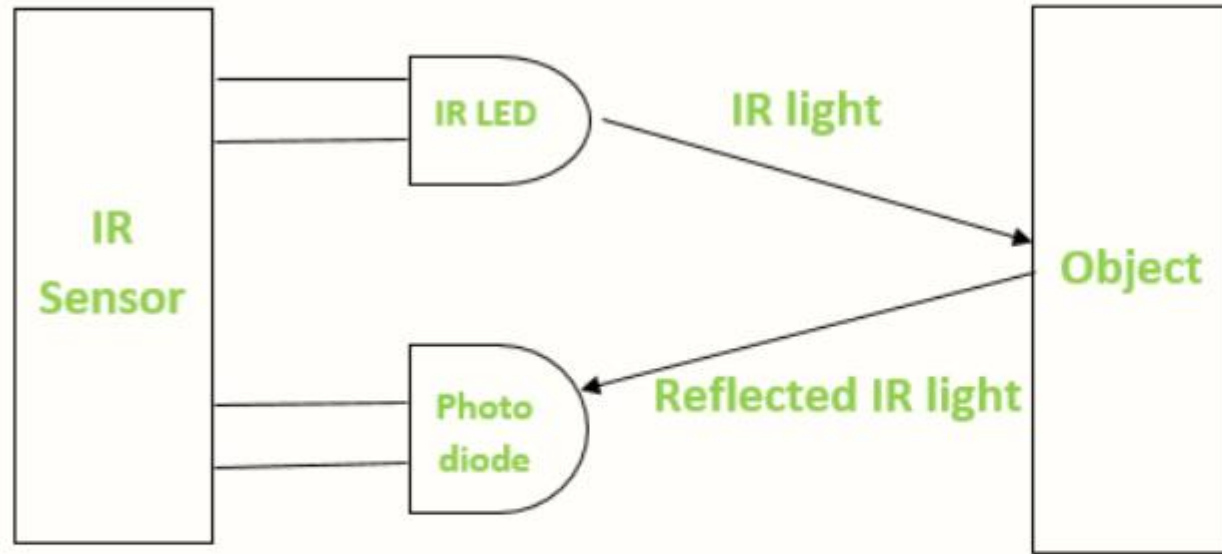
Infrared receivers or infrared sensors detect the radiation from an IR transmitter.

IR receivers come in the form of photodiodes.

Infrared Photodiodes are different from normal photo diodes as they detect only infrared radiation. Below image shows the picture of an IR receiver or a photodiode,

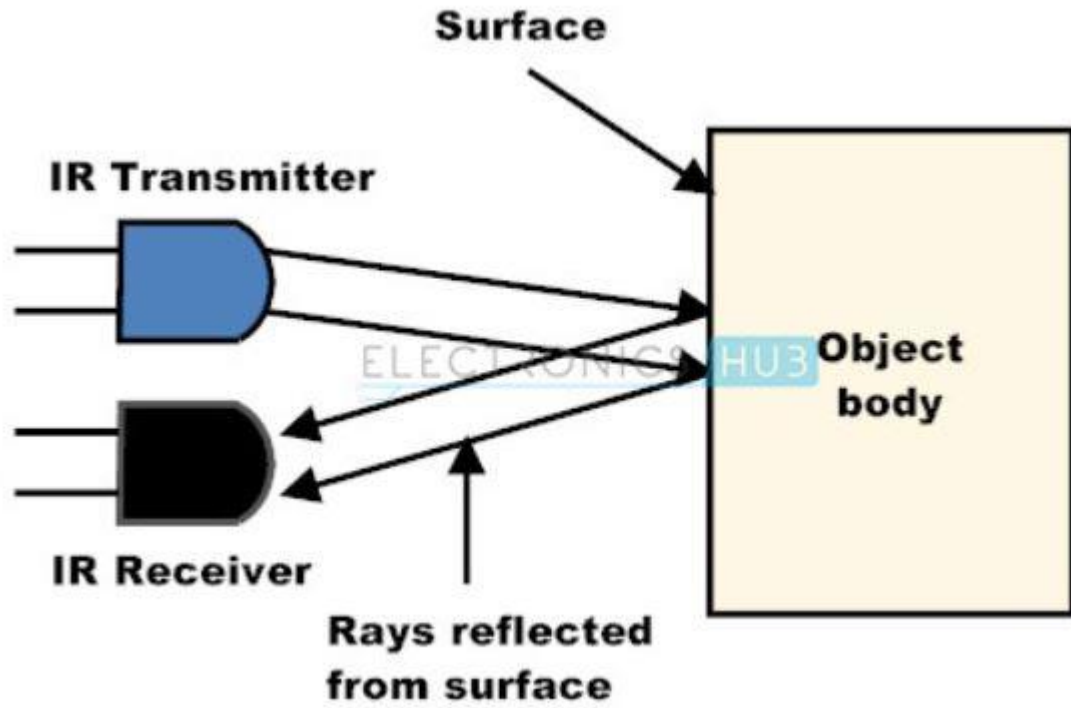


# IR Sensor Working Principle

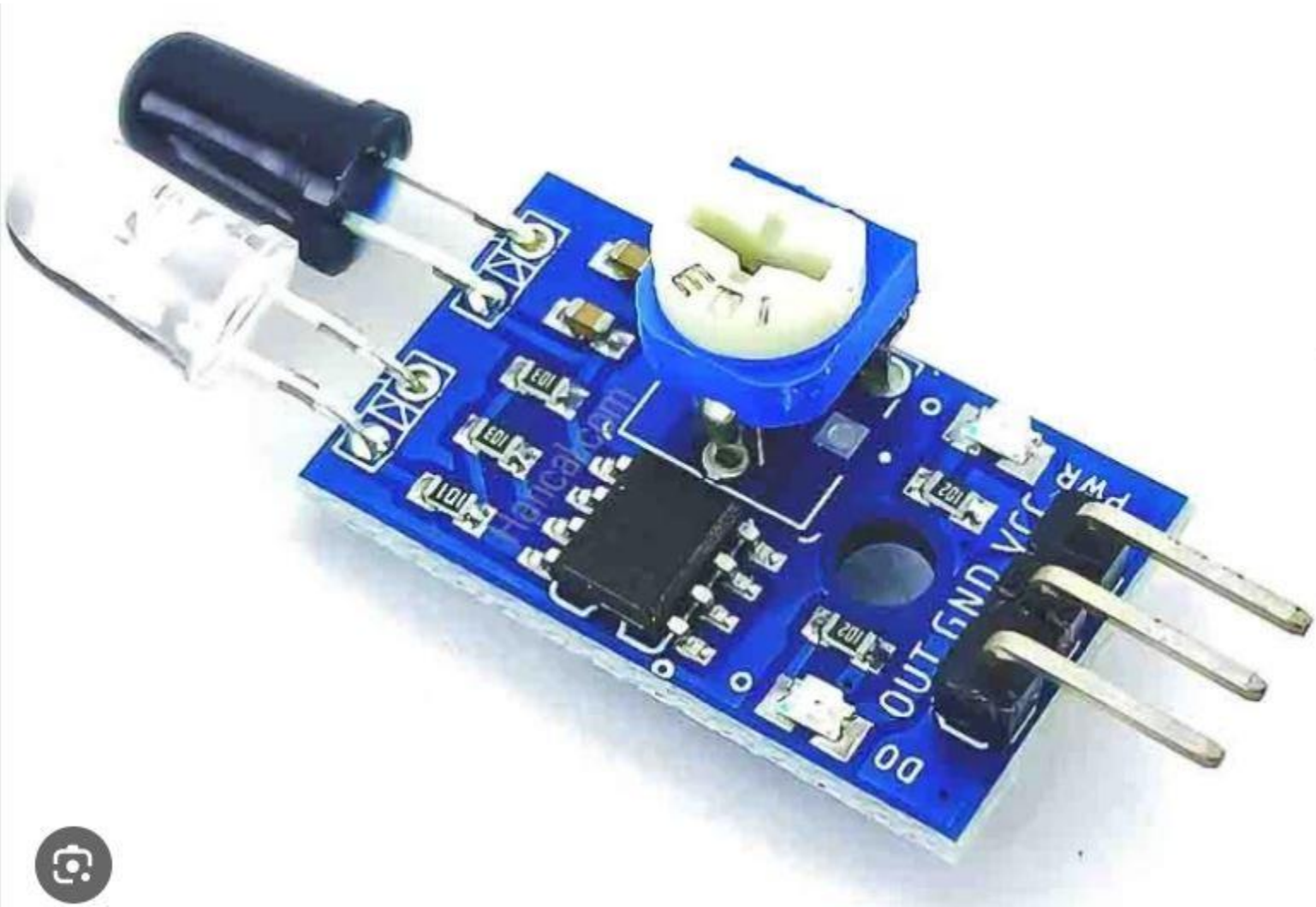


When the IR transmitter emits radiation, it reaches the object and some of the radiation reflects back to the IR receiver. Based on the intensity of the reception by the IR receiver, the output of the **sensor** defines.

# IR Sensor



# IR Sensor



# InfraRed (IR) Sensor

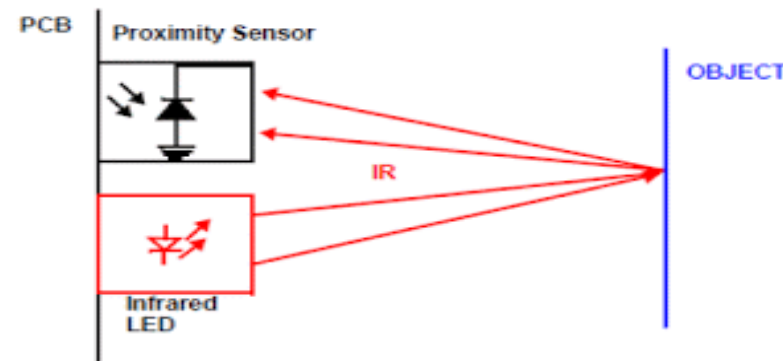


## Features-

- Can be used for obstacle sensing etc
- Input Voltage: 5V DC
- Comes with an easy to use digital output
- Can be used for wireless communication and sensing IR remote signals

IR Sensor have three Pins

1. VCC = +5V DC
2. GND
3. D0 or OUT (Digital Output)



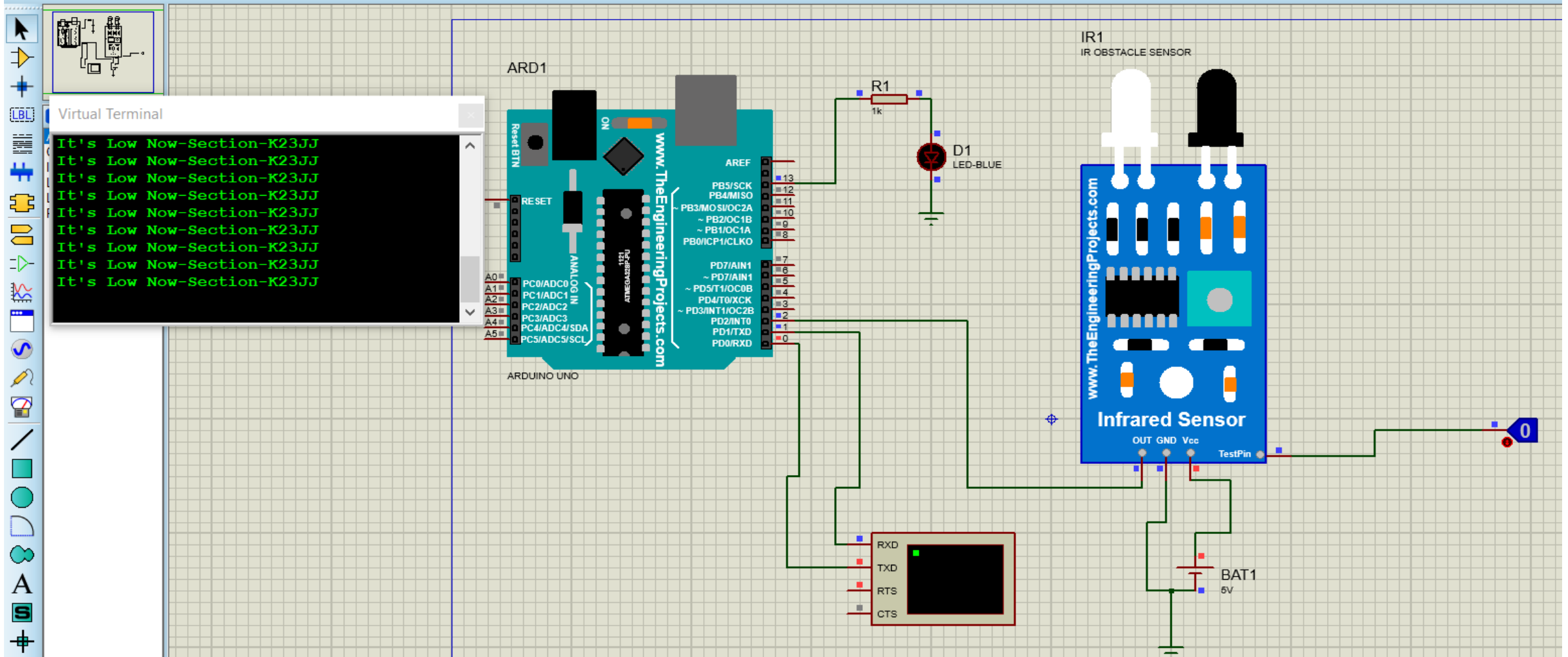
# IR Sensor(1 of the experiment based upon it)

IRSensor20march1244 - Proteus 8 Professional - Schematic Capture

File Edit View Tool Design Graph Debug Library Template System Help



Schematic Capture X





# IR Sensor(Code)

IRsensor | Arduino 1.8.19

File Edit Sketch Tools Help



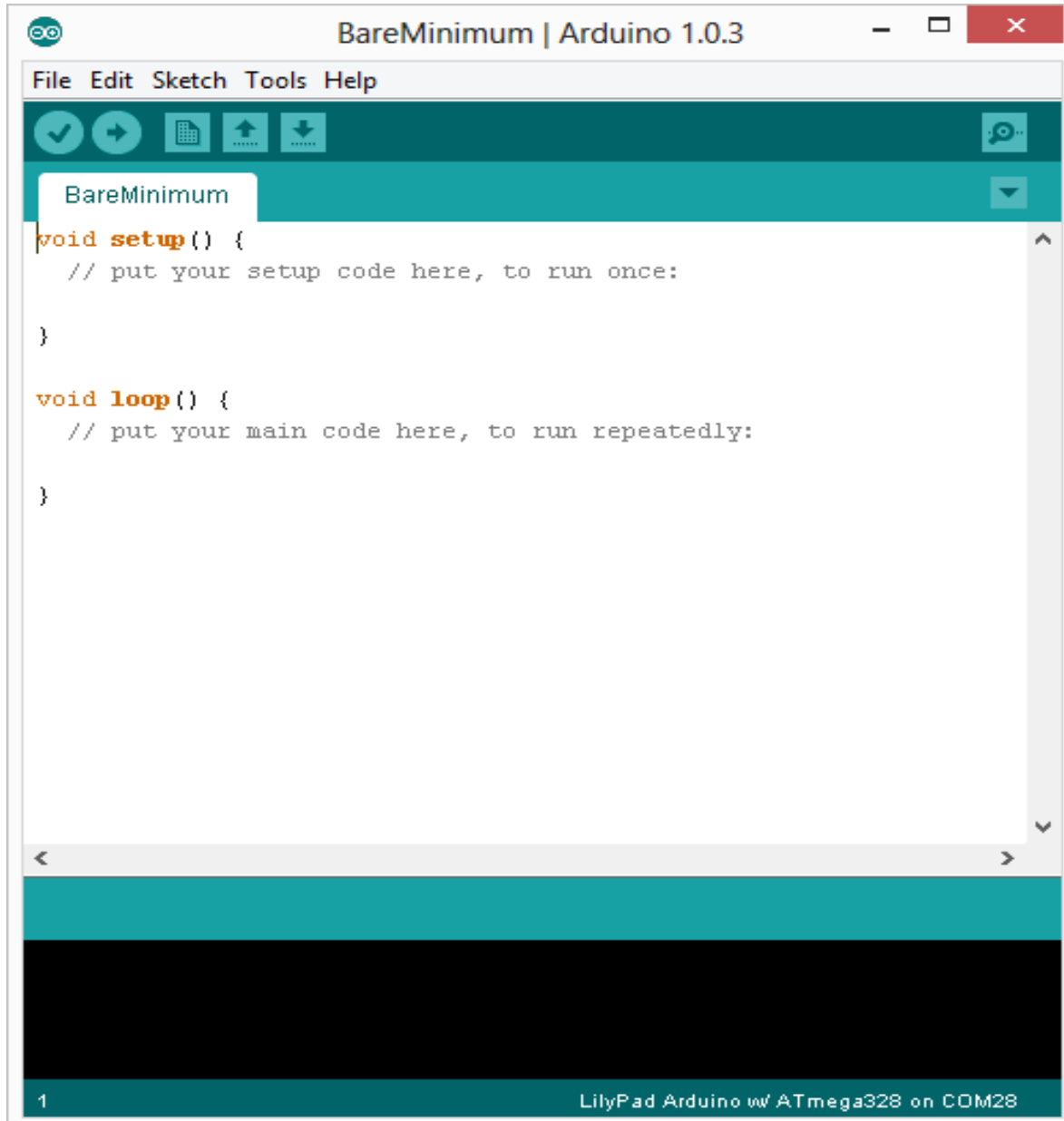
IRsensor

```
const int ProxSensor=2;
void setup()
{
    Serial.begin(9600);
    pinMode(13, OUTPUT);
    pinMode(ProxSensor, INPUT);
}
void loop()
{
    if(digitalRead(ProxSensor)==HIGH)
    {
        digitalWrite(13, HIGH);
        Serial.println("It's High Now-Name:Rahul & Roll Number:36");
    }
    else
    {
        digitalWrite(13, LOW);
        Serial.println("It's Low Now-Section-K23JJ");
    }
    delay(100);
}
```

**Sketch→Export Compiled Binary---  
for Hex file of Proteus Arduino**

we have the command 'Serial. begin(9600)'. This starts serial communication, so that the Arduino can send out commands through the USB connection. The value 9600 is called the 'baud rate' of the connection.

# Arduino IDE (Cont..)



Two required functions /  
methods / routines:

```
void setup()  
{  
    // runs once  
}
```

```
void loop()  
{  
    // repeats  
}
```

**Proteus**

# Proteus Software



# Proteus Software

New Project Wizard: Start

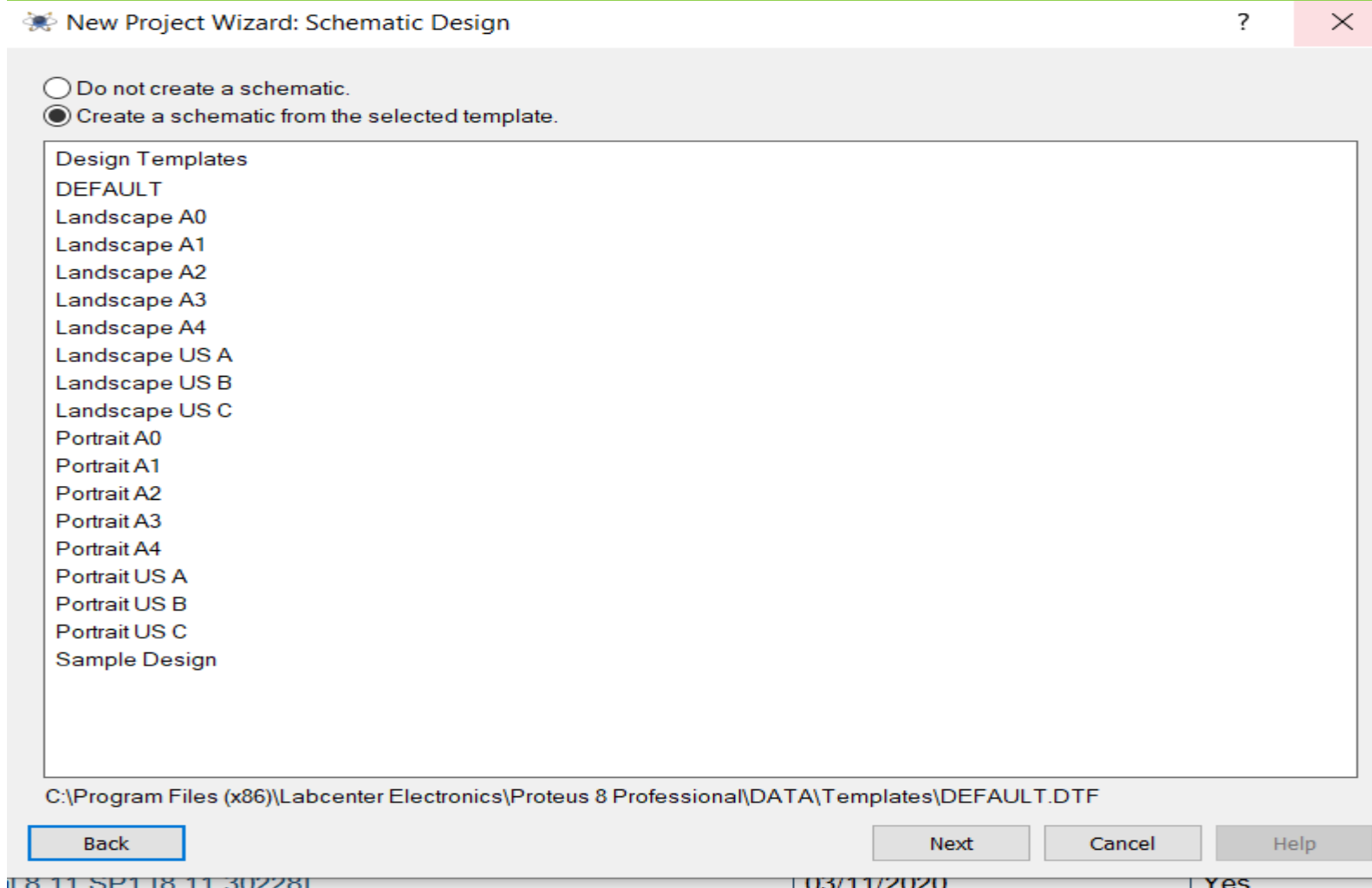
Project Name

Name

Path

☒ New Project ☐ From Development Board ☐ Blank Project

# Proteus Software



# Proteus Software

New Project Wizard: PCB Layout

☒ Do not create a PCB layout.  
☐ Create a PCB layout from the selected template.

Layout Templates

- Arduino MEGA 2560 rev3
- Arduino UNO rev3
- DEFAULT
- Double Eurocard (2 Layer)
- Double Eurocard (4 Layer)
- Extended Double Eurocard (2 Layer)
- Extended Double Eurocard (4 Layer)
- Generic Eight Layer 1.6mm (5 x Signal, 3 x Plane)
- Generic Four Layer 1.6mm (2 x Signal, 2 x Plane)
- Generic Single Layer
- Generic Six Layer 1.6mm (4 x Signal, 2 x Plane)
- PANEL
- Single Eurocard (2 Layer)
- Single Eurocard (4 Layer)
- Single Eurocard with Connector

C:\Program Files (x86)\Labcenter Electronics\Proteus 8 Professional\DATA\Templates\DEFAULT.LTF

Back Next Cancel Help

# Proteus Software

New Project Wizard: Firmware

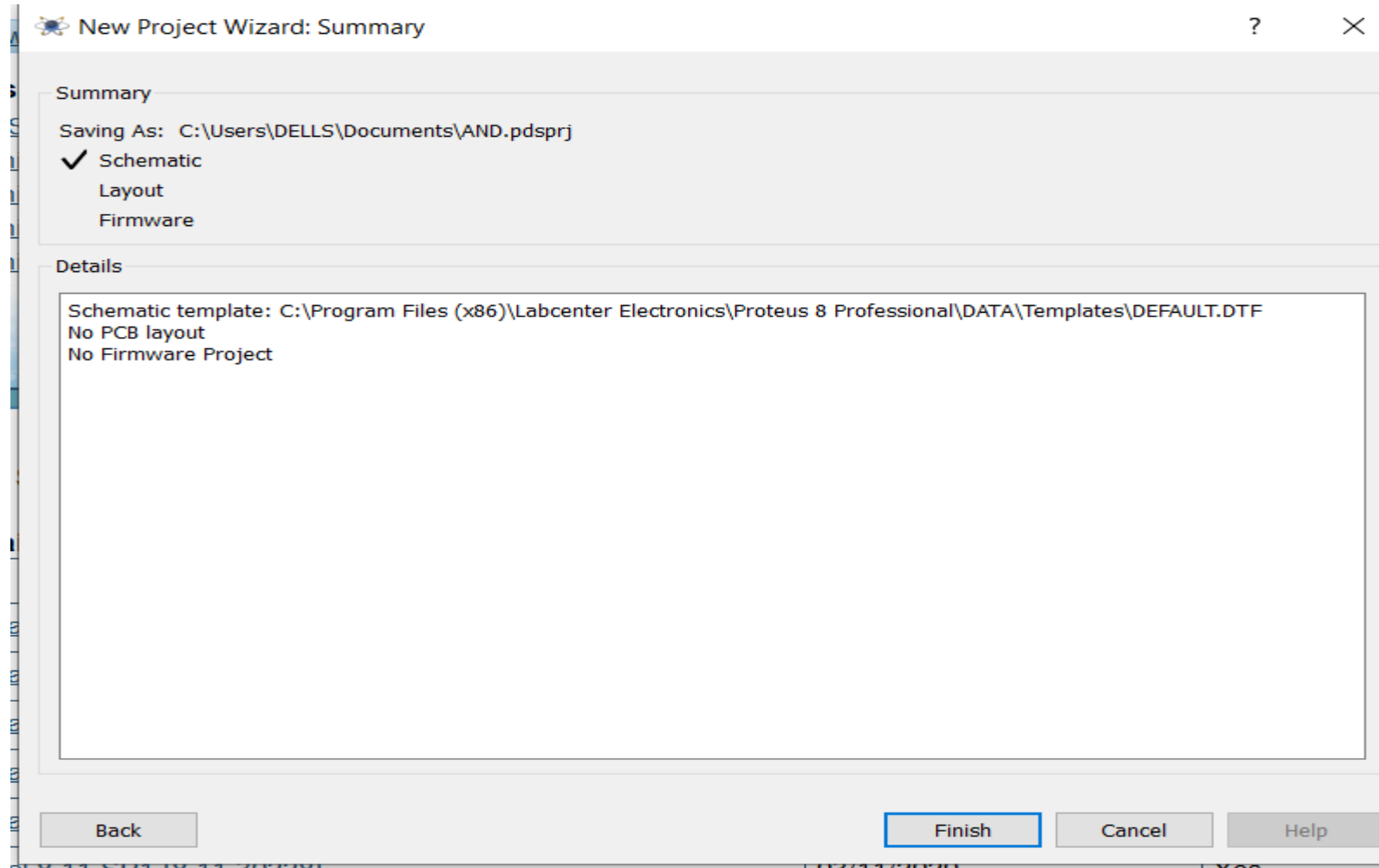
☒ No Firmware Project  
☐ Create Firmware Project  
☐ Create Flowchart Project

Family   
Controller   
Compiler

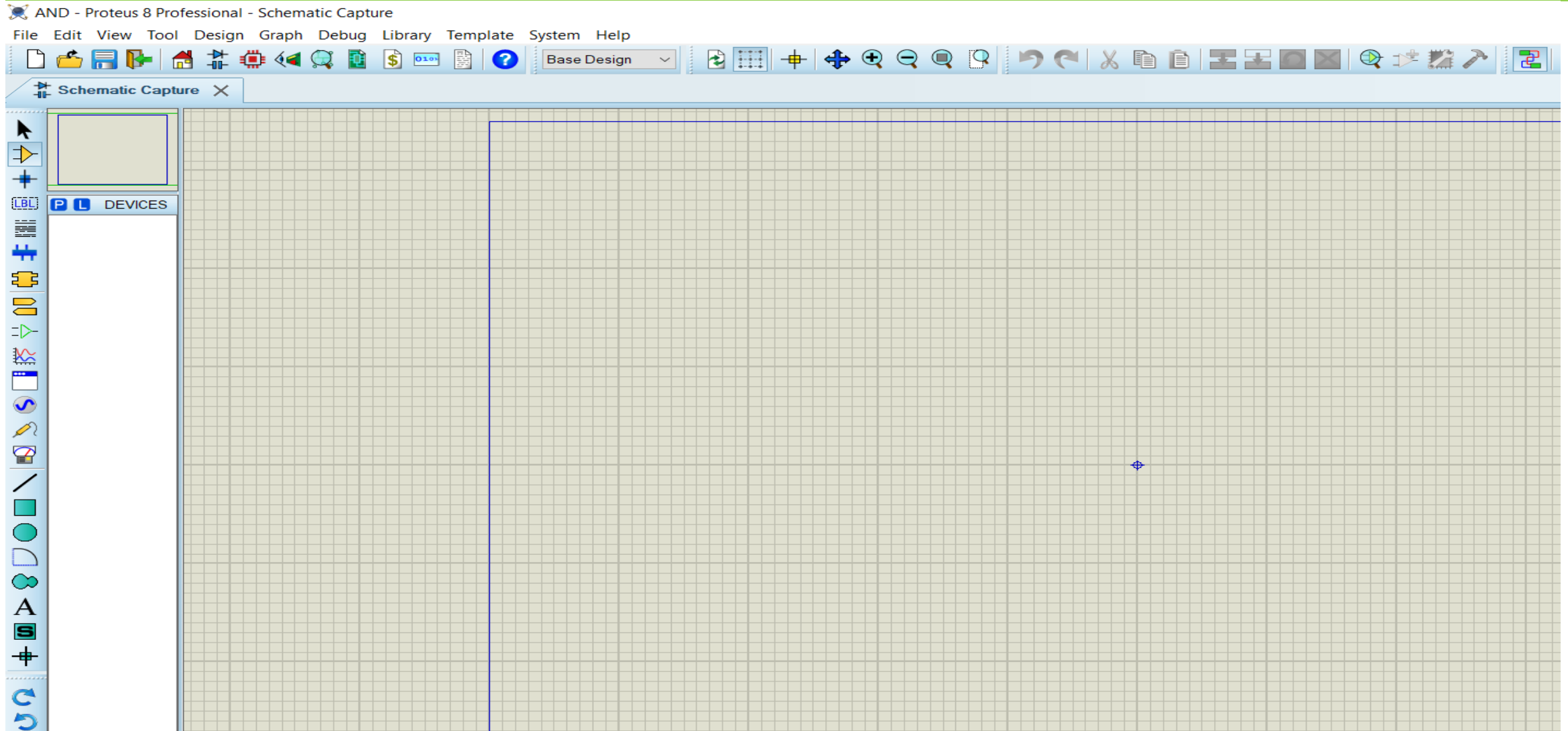
Create Quick Start Files ☐  
Create Peripherals ☐



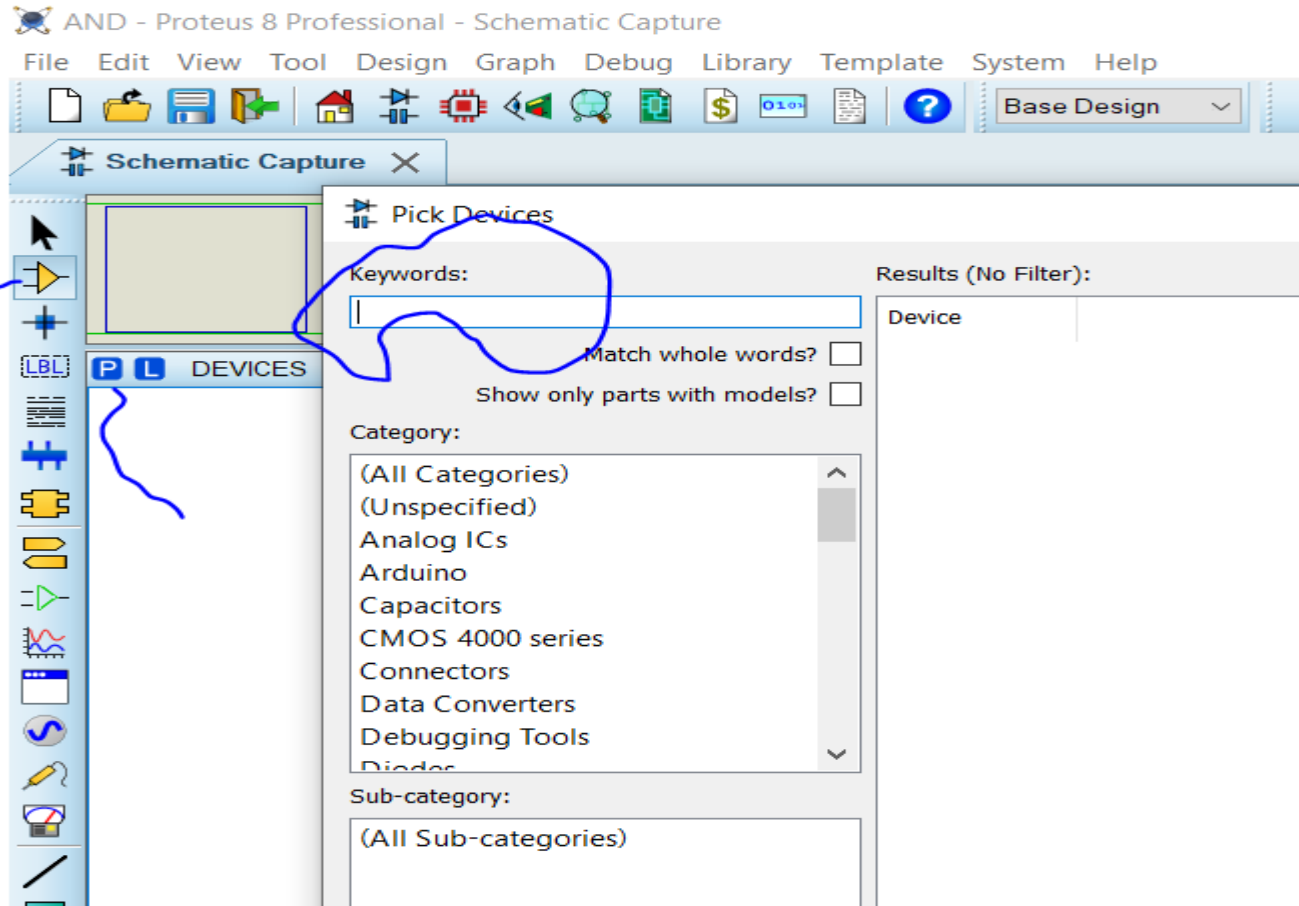
# Proteus Software



# Proteus Software



# Proteus Software



# Proteus Software

Pick Devices

?

Keywords:

Showing local results: 1659

and

Match whole words? ☐

Show only parts with models? ☐

Category:

(All Categories)

Analog ICs

CMOS 4000 series

Connectors

Data Converters

Diodes

Electromechanical

Inductors

Laplace Primitives

Memory ICs

Sub-category:

(All Sub-categories)

2nd Order

8051 Family

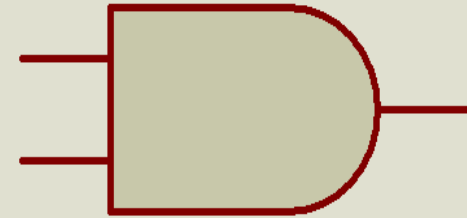
A/D Converters

Amplifiers

| Device    | Library  | Description                                                                                      |
|-----------|----------|--------------------------------------------------------------------------------------------------|
| ADC1251   | NATDAC   | Self-Calibrating 12-Bit Plus Sign ADC with Sample and Hold (12-bit, 8us conversion)              |
| ADC12662  | NATDAC   | 12-Bit, 1.5 MHz, 200 mW ADC with Input Multiplexer and Sample/Hold (12-bit, 0.58us conversion)   |
| ADC12L030 | NATDAC   | 3.3V Self-Calibrating 12-Bit Plus Sign Serial I/O ADC with MUX and Sample/Hold (12-bit, 8.80u... |
| ADC12L032 | NATDAC   | 3.3V Self-Calibrating 12-Bit Plus Sign Serial I/O ADC with MUX and Sample/Hold (12-bit, 8.80u... |
| ADC12L034 | NATDAC   | 3.3V Self-Calibrating 12-Bit Plus Sign Serial I/O ADC with MUX and Sample/Hold (12-bit, 8.80u... |
| ADC12L038 | NATDAC   | 3.3V Self-Calibrating 12-Bit Plus Sign Serial I/O ADC with MUX and Sample/Hold (12-bit, 8.80u... |
| ADC_10    | DSIMMDLS | Analog-to-Digital Converter (ADC) Primitive Model With Clock, Hold, References And Output E...   |
| ADC_12    | DSIMMDLS | Analog-to-Digital Converter (ADC) Primitive Model With Clock, Hold, References And Output E...   |
| ADC_16    | DSIMMDLS | Analog-to-Digital Converter (ADC) Primitive Model With Clock, Hold, References And Output E...   |
| ADC_8     | DSIMMDLS | Analog-to-Digital Converter (ADC) Primitive Model With Clock, Hold, References And Output E...   |
| AND       | ACTIVE   | Simple Two Input AND Gate                                                                        |
| AND_2     | DSIMMDLS | AND Gate Digital Primitive Model                                                                 |
| AND_2.DM  | DSIMMDLS | AND Gate Digital Primitive Model                                                                 |
| AND_3     | DSIMMDLS | AND Gate Digital Primitive Model                                                                 |
| AND_4     | DSIMMDLS | AND Gate Digital Primitive Model                                                                 |
| AND_5     | DSIMMDLS | AND Gate Digital Primitive Model                                                                 |
| AND_7     | DSIMMDLS | AND Gate Digital Primitive Model                                                                 |
| AND_8     | DSIMMDLS | AND Gate Digital Primitive Model                                                                 |
| AT24C1024 | I2CMEMS  | 1M-bit 138,072x8 I2C serial EEPROM memory with WPT and address select (400kHz @ 2.7V/10...       |
| AT24C1024 | I2CMEMS  | 1M-bit 138,072x8 I2C serial EEPROM memory with WPT and address select (400kHz @ 2.7V/10...       |

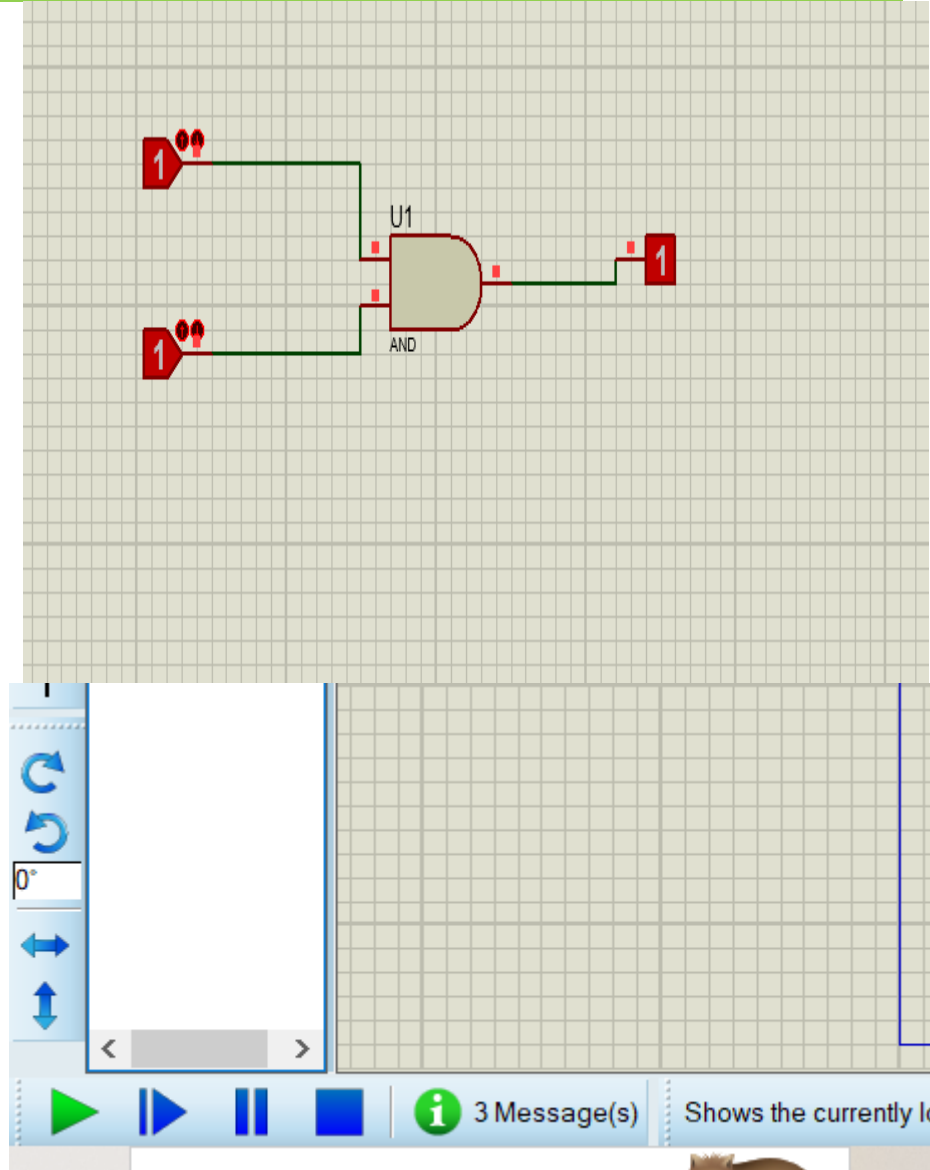
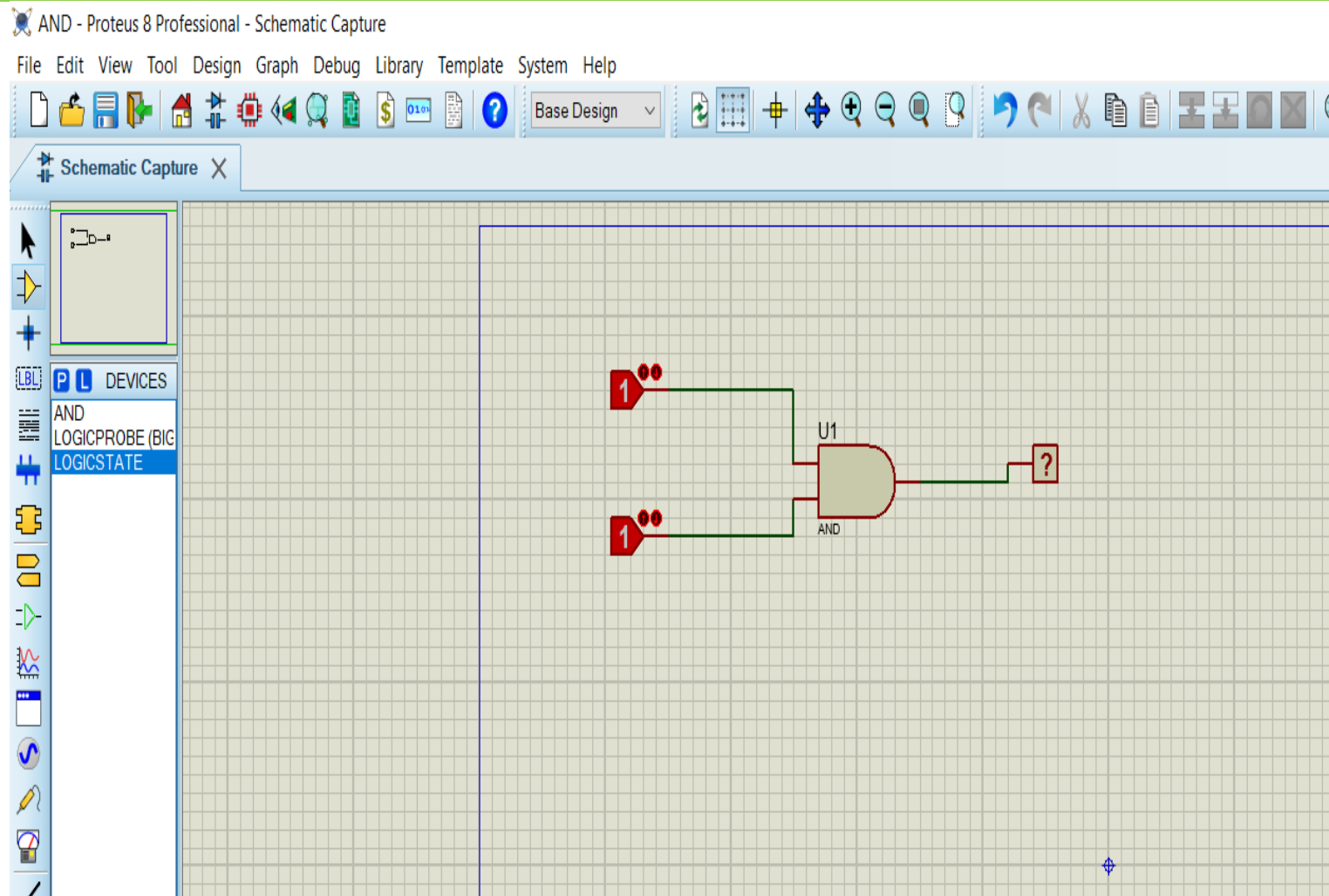
Preview

Digital Primitive [AND\_2]



PCB Preview

# Proteus Software



C:\Users\DELLS\Desktop\ProteusSoftware\1. And Gates